Lecture 3

Recap

1. The trig. form of the Fourier series of a T-periodic fⁿ f is

$$f(t) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos n\Omega t + b_n \sin n\Omega t)$$

where

$$a_n = c_n + c_{-n} = \frac{2}{T} \int_0^T f(t) \cos n\Omega t \, dt$$

$$b_n = i(c_n - c_{-n}) = \frac{2}{T} \int_0^T f(t) \sin n\Omega t \, dt$$

2. Simplifications for even/odd fⁿs:
   f even $\Rightarrow b_n = 0$
   f odd $\Rightarrow a_n = 0$
3. Can extend a function on $[0, T/2)$ to an odd/even fⁿ on $[-T/2, T/2)$.
   even extension $\Rightarrow$ cosine half-range series
   odd extension $\Rightarrow$ sine half-range series

Some spaces

- $L^1([0,T]) = L^1(\mathbb{T}) = \{ f \text{ T-periodic} : \int_0^T |f(t)| \, dt < \infty \}$
- $C^n = \{ f \text{ T-periodic} : f^{(k)} \text{ exists \& is continuous for } 0 \le k \le n \}$

1.5. Decay rate of Fourier coefficients

The aim of today's lecture is to understand when '$\sim$' can be replaced by '$=$' for a given Fourier series.

Recall that infinite series can display several behaviours:

- $\sum_{n=1}^{\infty} 1/n$ ← divergent
- $\sum_{n=1}^{\infty} (-1)^n/n = \log(2)$ ← conditionally convergent
  - ↪ $\sum_{n=1}^{\infty} a_n$ is finite but $\sum_{n=1}^{\infty} |a_n|$ isn't
- $\sum_{n=1}^{\infty} (-1)^n/n^2 = -\pi^2/12$ ← absolutely convergent
  - ↪ Both $\sum_{n=1}^{\infty} |a_n|$ & $\sum_{n=1}^{\infty} a_n$ are finite.

We first want to understand how c_n decays.

Prop 1.1 For $f \in L^1([0,T])$ w/ Fourier coeffs. c_n:

(i) $|c_n| \leq \frac{1}{T}\int_0^T |f(t)|\,dt$ (ii) $c_n \to 0$ as $n \to \infty$.

Proof: (i) The triangle inequality gives

$$|c_n| \leq \frac{1}{T}\int_0^T |f(t)e^{-in\Omega t}|\,dt$$
$$= \frac{1}{T}\int_0^T |f(t)|\,dt < \infty.$$

(ii) This is the Riemann-Lebesgue Lemma & we do not need to understand the proof for this course.

This result tells us that $|c_n|$ are bounded above & decay as $n \to \infty$, but this doesn't guarantee convergence.

⇝ Think about the harmonic series!

Now we put more structure on f to get a stronger result.

Prop 1.2: Let $f \in C^k$ be T-periodic. There exists an $M > 0$ such that

$$|c_n| \leq \frac{M}{|n|^k}, \quad \forall n \neq 0.$$

Proof: By hypothesis, $f^{(k)}$ is T-periodic & continuous. This implies that $f^{(k)} \in L^1([0,T])$, so we are free to compute the Fourier coeffs of it:

$$d_n = \frac{1}{T}\int_0^T f^{(k)}(t)\,e^{-in\Omega t}\,dt$$

Now integrate by parts

$$d_n = \frac{1}{T}\left[f^{(k-1)}(t)\,e^{-in\Omega t}\right]_0^T + \frac{i\Omega n}{T}\int_0^T e^{-in\Omega t} f^{(k-1)}(t)\,dt$$

$$= \frac{1}{T}\left(f^{(k-1)}(T) - f^{(k-1)}(0)\right) + \frac{i\Omega n}{T}\int_0^T f^{(k-1)}(t)e^{-in\Omega t}\,dt$$

$f^{(k-1)}(T) = f^{(k-1)}(0)$

$$= \frac{i\Omega n}{T}\int_0^T f^{(k-1)}(t)e^{-in\Omega t}\,dt.$$

We can repeat this process until we get

$$d_n = \frac{(i\Omega n)^k}{T}\int_0^T f(t)\,e^{-in\Omega t}\,dt$$

$$= (i\Omega n)^k\, c_n.$$

$$\Rightarrow |c_n| = |d_n| / (\Omega |n|)^k.$$

Using part (i) of the previous prop., we know that $|d_n|$ is bounded above

$$|d_n| \leq \frac{1}{T}\int_0^T |f^{(k)}(t)|\,dt \doteq M\Omega^k.$$

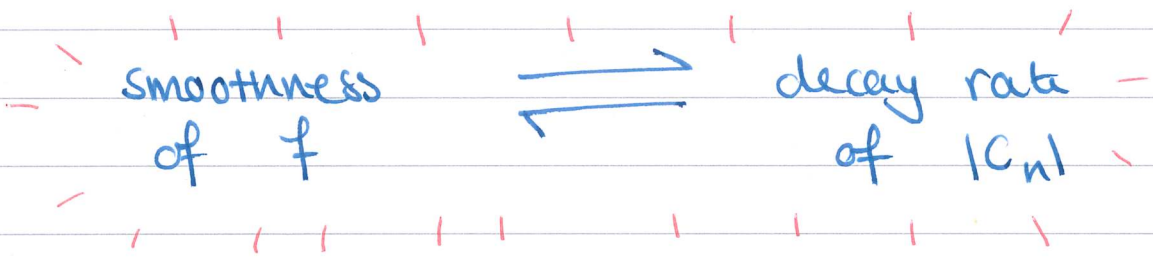
Therefore

$$|c_n| \leq \frac{M}{|n|^k},$$

where

$$M = \frac{1}{T\Omega^k}\int_0^T |f^{(k)}(t)|\,dt.$$

This result gives us the general idea



Using this we can establish a class of f's whose Fourier series are absolutely convergent.

Let f be T-periodic & twice continuously differentiable. Then

$$\left| \sum_{n=-\infty}^{\infty} c_n e^{in\Omega t} \right| \leq |c_0| + \sum_{n=1}^{\infty} (|c_n| + |c_{-n}|)$$

$$= |c_0| + \sum_{n=1}^{\infty} \frac{2M}{n^2},$$

where $M = \frac{1}{T\Omega^2} \int_0^T |f^{(2)}(t)| dt$. Since $\sum_{n=1}^{\infty} 1/n^2$ is convergent, we have bounded the left-hand side, so the Fourier series of $f(t)$ is absolutely convergent.

1.6 Convergence of Fourier series

The last result establishes when absolute convergence occurs, but if f isn't twice continuously differentiable, this doesn't mean it Fourier series is divergent — it could be conditionally convergent.

Now we want to compare the value of the sum to the values of f. We define the Nth partial sum $S_N(t)$ to be

$$S_N(t) = \sum_{n=-N}^{N} c_n e^{inlt},$$

for a given set of Fourier coefficients c_n.

Question: $\lim_{N\to\infty} S_N(t) = ?$

Recall that the left & right limits of a f^n are

$$f(c^-) = \lim_{t\to c^-} f(t), \quad f(c^+) = \lim_{t\to c^+} f(t)$$

approach from below $t<c$; approach from above $t>c$

If f is continuous at $t=c$, then

$$f(c^-) = f(c) = f(c^+).$$

But f is discontinuous, they do not match.

Similarly, the left derivative is

$$f_-'(c) = \lim_{h\to 0^-} \frac{f(c+h) - f(c^-)}{h}$$

while the right derivative is

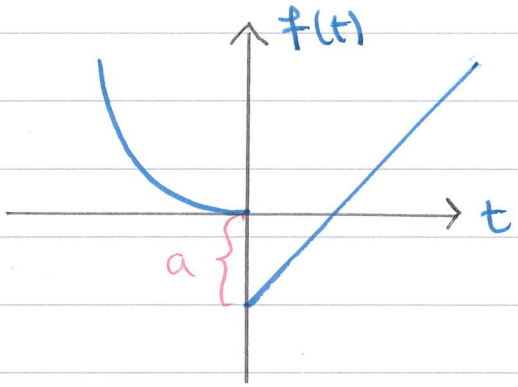
$$f_+'(c) = \lim_{h\to 0^+} \frac{f(c+h) - f(c^+)}{h}.$$

If f is continuous & differentiable, these match & we can unambiguously define $f'(c)$.

Example 1.8

Consider the fn

$$f(t) = \begin{cases} t^2, & t < 0 \\ t - a, & t \geq 0. \end{cases}$$



We want to determine the left/right limits & derivatives at $t=0$. Clearly,

$$f(0^-) = \lim_{t \to 0^-} t^2 = 0$$

$$f(0^+) = \lim_{t \to 0^+} (t - a) = -a.$$

$\Longrightarrow$ f continuous iff $a=0$.

Now we can compute the derivatives:

$$f'_-(0) = \lim_{h \to 0^-} \frac{(0+h)^2 - 0}{h}$$

$$= \lim_{h \to 0^-} h = 0$$

$$f'_+(0) = \lim_{h \to 0^+} \frac{(0+h) - a + a}{h}$$

$$= \lim_{h \to 0^+} \frac{h}{h} = 1.$$

$\Longrightarrow$ Never differentiable, even if $a=0$.

Thm 1.1: Let f be a T-periodic f^n such that $f'_+(t)$ & $f'_-(t)$ exist everywhere. Then the partial sums $S_N(t)$ converge pointwise to

$$\lim_{N\to\infty} S_N(t) = \frac{f(t^+) + f(t^-)}{2}$$

$$\text{by def}^n \uparrow \quad = \sum_{n=-\infty}^{\infty} c_n e^{in\Omega t}$$

for every $t \in \mathbb{R}$. ← (Also holds for trig form via Euler's formula)

This is Dirichlet's criterion. At points of continuity, it lets us replace 'N' with '$=$' so long the left & right limits exist. At points of discontinuity, it equals the average of the left & right limits

↪ This happened w/ the sawtooth!

For the sawtooth f^n, we saw it was discontinuous at $t = \pi$. There

$$f(\pi^-) = \pi, \quad f(\pi^+) = -\pi$$

$$\Rightarrow \frac{f(\pi^-) + f(\pi^+)}{2} = 0.$$

↪ we saw the Fourier series also equalled 0 there!

Example 1.9 We're going to use Dirichlet's criterion on both $f(t) = t^2$ on $[-1,1)$ & $f(t) = |\sin t|$ to derive results about series. ← Both examples were done in Lecture 2.

(i) we saw that

$$t^2 \sim \frac{1}{3} + \sum_{n=1}^{\infty} \frac{4(-1)^n}{\pi^2 n^2} \cos(n\pi t),$$

for $t \in [-1,1)$. Note that $t=0$ is a point of continuity, so '$\sim$' can be replaced by '$=$' there. Doing so gives

and differentiable there too!

$$0 = \frac{1}{3} + \sum_{n=1}^{\infty} \frac{4(-1)^n}{\pi^2 n^2}$$

$$\Rightarrow \sum_{n=1}^{\infty} \frac{1}{n^2} = -\frac{\pi^2}{12}.$$

(ii) For $f(t) = |\sin t|$, we saw

$$|\sin t| \sim \frac{2}{\pi} - \frac{4}{\pi} \sum_{n=1}^{\infty} \frac{\cos(2nt)}{4n^2 - 1}.$$

While $t=0$ is a point of continuity, it is not differentiable there. Need to check $f'_{\pm}(0)$ exist.

Taylor expand →

$$f'_{-}(0) = \lim_{h \to 0^-} -\frac{\sin h}{h} = \lim_{h \to 0^-} \left[\frac{-h + h^3/6 + \dots}{h} \right]$$

$$= -1$$

$$f_+'(0) = \lim_{h \to 0^+} \frac{\sin h}{h} = 1,$$

also by Taylor expansion (or l'Hôpital's rule). Therefore, we can replace $\sim$ with $=$ at $t=0$:

$$0 = \frac{2}{\pi} - \frac{4}{\pi} \sum_{n=1}^{\infty} \frac{1}{4n^2-1}$$

$$\Rightarrow \sum_{n=1}^{\infty} \frac{1}{4n^2-1} = \frac{1}{2}.$$

When is this useful?

Beyond getting fun results about series, this gives us nice consistency checks for our answers.

1. Check f satisfies the Dirichlet criterion.

2. Obtain a series result

3. If correct, the Fourier coefficients are consistent at that point.